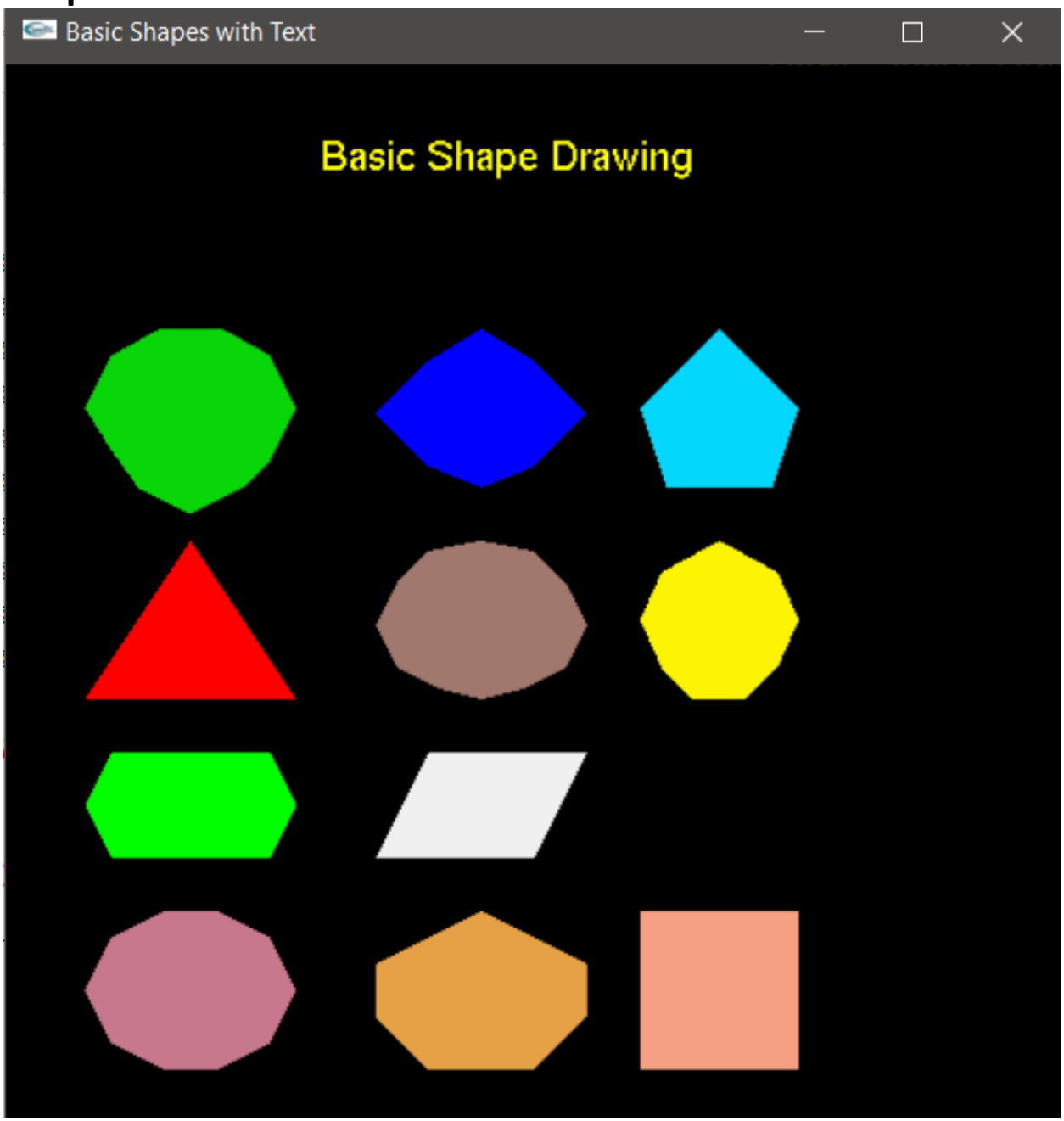
# Source Code:

```
#include <windows.h>
#include <GL/gl.h>
#include <GL/glut.h>

// □ Simple text drawing function
void drawText(float x, float y, char *text)
{
   glRasterPos2f(x, y);  // Set text position
   for (int i = 0; text[i] != '\0'; i++)
   {
      glutBitmapCharacter(GLUT_BITMAP_HELVETICA_18, text[i]);  // Print each character
   }
}

// ◎ Display callback function
void display(void)
{
   glClear(GL_COLOR_BUFFER_BIT); // Clear screen

   // □ Draw yellow title text
   glColor3f(1.0, 1.0, 0.0);
   drawText(30, 90, "Basic Shape Drawing");


 /*  //single Point Drawing
   glPointSize(10.0f);              // MUST be float
   glColor3ub(255, 255, 255);        // Correct integer color (0–255)
   glBegin(GL_POINTS);
      glVertex2f(10.0f, 90.0f);       // Visible with default projection
   glEnd();


   glBegin(GL_LINES);
         glVertex2f(10.0f, 50.0f); // Start point
         glVertex2f(40.0f, 50.0f); // End point
      glEnd();

   //Not filled
   glBegin(GL_LINE_LOOP);
         glVertex2f(50.0f, 10.0f); // Bottom left
         glVertex2f(80.0f, 10.0f); // Bottom right
         glVertex2f(65.0f, 40.0f); // Top center
      glEnd();

 */
```

```
//10 edge
   glColor3ub(200, 120, 140);
   glBegin(GL_POLYGON);
   glVertex2f(10, 7.5);
   glVertex2f(7.5, 12.5);
   glVertex2f(10, 17.5);
   glVertex2f(15, 20);
   glVertex2f(20, 20);
   glVertex2f(25, 17.5);
   glVertex2f(27.5, 12.5);
   glVertex2f(25, 7.5);
   glVertex2f(20, 5);
   glVertex2f(15, 5);

   glEnd();

//hexagon
 glColor3f(0.0f, 1.0f, 0.0f);
 glBegin(GL_POLYGON);
 glVertex2f(10, 25);
 glVertex2f(7.5, 30);
 glVertex2f(10, 35);
 glVertex2f(25, 35);
 glVertex2f(27.5, 30);
 glVertex2f(25, 25);

 glEnd();

 //triangle
 glColor3f(1.0f, 0.0f, 0.0f);
 glBegin(GL_TRIANGLES);
 glVertex2f(7.5, 40);
 glVertex2f(17.5, 55);
 glVertex2f(27.5, 40);

 glEnd();

//11edges
glColor3ub(9, 214, 9);
glBegin(GL_POLYGON);
glVertex2f(12.5, 60);
glVertex2f(10, 63.5);
glVertex2f(7.5, 67.5);
glVertex2f(10, 72.5);
glVertex2f(14.5, 75);
glVertex2f(20.5, 75);
glVertex2f(25, 72.5);
glVertex2f(27.5, 67.5);
```

```
glVertex2f(25, 62.5);
glVertex2f(22.5, 60);
glVertex2f(17.5,57.5);
glEnd();

//Heptagon
glColor3ub(230, 160, 70);
glBegin(GL_POLYGON);
glVertex2f(40, 5);
glVertex2f(35, 10);
glVertex2f(35, 15);
glVertex2f(45, 20);
glVertex2f(55, 15);
glVertex2f(55, 10);
glVertex2f(50, 5);

glEnd();

//Parralelogram
glColor3ub(240,240,240);
glBegin(GL_QUADS);
glVertex2f(35, 25);
glVertex2f(40, 35);
glVertex2f(55, 35);
glVertex2f(50, 25);

glEnd();

//dodecagon
glColor3ub(160, 120, 110);
glBegin(GL_POLYGON);
glVertex2f(45, 40);
glVertex2f(41, 41);
glVertex2f(37, 43);
glVertex2f(35, 47);
glVertex2f(37, 51);
glVertex2f(40, 54);
glVertex2f(45, 55);
glVertex2f(50, 54);
glVertex2f(53, 51);
glVertex2f(55, 47);
glVertex2f(53, 43);
glVertex2f(49, 41);

glEnd();

//octagon
glColor3f(0.0f, 0.0f, 1.0f);
```

```
glBegin(GL_POLYGON);
glVertex2f(45, 60);
glVertex2f(40, 62);
glVertex2f(35, 67);
glVertex2f(40, 72);
glVertex2f(45, 75);
glVertex2f(50, 72);
glVertex2f(55, 67);
glVertex2f(50, 62);
glEnd();


//rectangle
glColor3ub(245, 160, 130);
glBegin(GL_QUADS);
glVertex2f(60, 5);
glVertex2f(60, 20);
glVertex2f(75, 20);
glVertex2f(75, 5);

glEnd();
//nonagon
glColor3ub(252, 244, 3);
glBegin(GL_POLYGON);
glVertex2f(70, 40);
glVertex2f(65, 40);
glVertex2f(62, 43);
glVertex2f(60, 47.5);
glVertex2f(62, 52);
glVertex2f(67.5, 55);
glVertex2f(73, 52);
glVertex2f(75, 47.5);
glVertex2f(73, 43);

glEnd();

//PENTAGON
glColor3ub(3, 215, 252);
glBegin(GL_POLYGON);
glVertex2f(62.5, 60);
glVertex2f(60, 67.5);
glVertex2f(67.5, 75);
glVertex2f(75, 67.5);
glVertex2f(72.5, 60);

glEnd();
```

```
 /*glColor3f(1.0f, 0.0f, 0.0f);
 drawText(8, 30, "Triangle");
 glEnd();
*/

   glFlush(); // Render everything
}

// □ Initialize OpenGL settings
void init(void)
{
   glClearColor(0.0, 0.0, 0.0, 0.0);  // Background color = black

   glMatrixMode(GL_PROJECTION);
   glLoadIdentity();

   glOrtho(0.0, 100.0, 0.0, 100.0, -1.0, 1.0);
}

// ● Main function
int main(int argc, char** argv)
{
   glutInit(&argc, argv);
   glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
   glutInitWindowSize(500, 500);
   glutInitWindowPosition(100, 100);
   glutCreateWindow("Basic Shapes with Text");

   init(); // Setup OpenGL
   glutDisplayFunc(display); // Register display callback
   glutMainLoop(); // Keep the window running
   return 0;
}
```

**Output:**